

grok time  $T_{\text{grok}}$  (steps)

- $p = 43$ :  $\alpha = 0.173$ ,  $R^2 = 0.987$
- $p = 59$ :  $\alpha = 0.140$ ,  $R^2 = 0.991$
- $p = 67$ :  $\alpha = 0.126$ ,  $R^2 = 0.982$
- $p = 97$ :  $\alpha = 0.094$ ,  $R^2 = 0.957$
- $p = 113$ :  $\alpha = 0.087$ ,  $R^2 = 0.959$

$10^4$

$10^3$

45

50

55

60

65

70

75

80

held weight norm  $\|W\|$

